VISCOSITY

22

Viscosity can be defined as the resistance of a fluid to flowing or changing shape. The more viscous a substance is, the slower it flows. Viscosity has important industrial relevance, especially when fluids are being transported through pipes. A group of students was set the task of measuring the viscosity of liquids. The practical procedure was set out as follows:

- Set up a clean and dry burette with a beaker underneath as shown in figure 1
- Place liquid into the burette to slightly above the zero mark
- Run out liquid until the meniscus rests on the zero mark
- Time how long it takes for 50 mL of the liquid to pour out of the burette into the beaker
- Rinse out the burette with the next liquid to be tested
- Repeat with the other liquids.



Figure 1

Table 1 shows the results collected from the six groups who carried out the experiment.

Liquid	Water	Milk	Ethanol	Cooking oil
Time to flow (sec)	22, 24, 27, 45, 29, 23	29, 30, 34, 19, 35, 31	25, 27, 31, 55, 33, 28	66, 72, 80, 122, 90, 71

Table 1

1 Use a diagram to help you explain what a *meniscus* is.

2 Describe how a burette is used to measure out a volume of liquid.

3 One group had problems with a burette. From each set of results, pick out the figure that is probably due to the faulty burette.

4 Leaving out the faulty readings, work out the average time for 50 mL of each liquid to flow from a burette.

5 List the liquids in order of increasing viscosity.

6 Why should the burette be rinsed out between experiments?

7 Name any essential apparatus not shown in figure 1 that is required to carry out the experiment.

Fluid particles have to move past each other in order to flow; therefore viscosity can be regarded as being due to the friction between moving molecules. Polar (electrically-charged) molecules are more likely to interact with each other than similar neutral molecules are. Viscosity changes with temperature; liquids flow better at higher temperatures, gases flow better at lower temperatures.

8 Describe what happens to the viscosity of each substance.

a liquids as the temperature is decreased

b gases as the temperature is increased

9 The students were asked to devise an alternative way of comparing the viscosity of liquids. One student included a diagram (see figure 2) but didn't give any instructions for carrying out the experiment. Write a set of instructions for comparing viscosities of liquids using the apparatus shown in figure 2. Include details of how you would ensure it was a fair test.

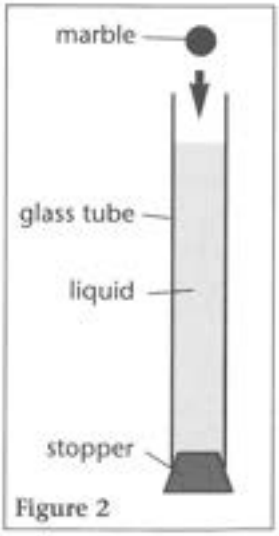


Figure 2

10 Name a viscous liquid not already mentioned on this page.

11 Mark each statement as true or false.

a Fluids are liquids and gases.

b The greater the interaction between fluid particles, the less the viscosity of that fluid.

c The rate of flow of a fluid is inversely proportional to its viscosity.

d Liquids are more viscous than gases.

e Viscosity of fluids is related to temperature.

f Polar molecules don't attract each other.

g Blood is more viscous than water.

h Viscosity can be defined as a substance's resistance to flow.

ISBN 978 1 92561 066 6
© Peter Gribben, Wendy St Germain and Blake Education 2005